RAJALAKSHMI ENGINEERING COLLEGE

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V2

Started on	Friday, 21 August 2026, 8:28 AM
State	Finished
Completed on	Friday, 21 August 2026, 8:41 AM
Time taken	12 mins 41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int find(int arr[],int left,int right, int x){
    if(left>=right){
        return 0;
    }
    int sum = arr[left]+arr[right];
    if(sum==x){
        printf("%d\n%d\n",arr[left],arr[right]);
        return 1;
    }
    else if(sum<x){
        return find(arr,left+1,right,x);
    }
    else{
        return find(arr,left,right-1,x);
    }
}

int main(){
    int n;
    if(scanf("%d",&n)!=1){
        return 0;
    }
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int x;
    scanf("%d",&x);
    if(!find(arr,0,n-1,x)){
        printf("No\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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